FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2025/2026

Subject	:	PROBABILITY & STATISTICAL DATA ANALYSIS (SECI 1143)
Task	:	Chapter 7 & 8
Due Date	:	**Week 15 (22 – 28 June 2026)**

INSTRUCTION:

i. This is a **GROUP** assignment (each group 3 – 4 members only). Please clearly write the group members **NAME & MATRIC NUMBER** in the front page of the submission.

ii. This assignment contributes to 5% of overall course marks.

iii. Only **HANDWRITTEN** submission is accepted:
   a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
   b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
   c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
   d. Round your **final answers** to **THREE** decimal places.
   e. Please scan/snapshot your work and save as a PDF file.

iv. Submission via eLearning – only **ONE** group member needs to submit on behalf of the group.

QUESTION 1 (15 MARKS)

A researcher is interested in examining whether there is a relationship between the amount of time students spend on social media and their academic performance. Social media platforms such as Instagram and TikTok have become an integral part of students' daily lives. While these platforms can facilitate communication and access to information, excessive use may reduce the time available for studying and completing academic tasks.

To investigate this issue, the researcher collects data from a sample of 10 undergraduate students at a university. Each student is asked to report the average number of hours they spend on social media per day. The researcher also obtains each student's current Cumulative Grade Point Average (CGPA) from university records, with the students' consent. Data from 10 students are collected as in Table 1.

Table 1

Student	Time	CGPA
A	3	3.70
B	2	3.77
C	3	3.80
D	6	3.55
E	12	2.57
F	5	3.52
G	4	3.60
H	8	3.14
I	4	3.65
J	10	2.80

a. Construct a scatter plot diagram, using the given data.

[5 marks]

b. Based on the plot, describe the direction, and strength of the relationship between the variables.

[2 marks]

c. Calculate the Pearson correlation coefficient, r. Show all calculations.

[6 marks]

d. Suppose a new observation is added for Student K with time spend on social media = 12 and GPA = 3.95. Predict the likely impact on the correlation coefficient and explain your reasoning.

[2 marks]

QUESTION 2 (20 MARKS)

The management of a university cafeteria is looking for ways to increase daily meal sales. One strategy they have recently implemented is sending promotional messages to students through the university's mobile application. The cafeteria manager believes that increasing the number of promotional messages may encourage more students to purchase meals from the cafeteria. To investigate this possibility, data were collected over eight consecutive days (Table 2). For each day, the number of promotional messages sent through the student app (X) and the corresponding number of meal sets sold were recorded (Y).

Table 2

Day	Promotional Messages (X)	Meal Sets Sold (Y)
1	1	120
2	3	151
3	5	177
4	4	160
5	3	132
6	2	127
7	7	200
8	10	224

The cafeteria manager would like to use simple linear regression to model the relationship between the number of promotional messages and the number of meal sets sold. The results of the analysis will help determine whether promotional messaging is an effective marketing strategy and assist in forecasting future sales.

Based on the data above, answer the following questions:

a. Calculate the least-squares regression equation, $\hat{y} = b_0 + b_1 x$. Show all calculations clearly.

[14 marks]

a. Using the regression equation obtained in part (a), predict the number of meal sets that would be sold if the cafeteria sends 7 promotional messages in a day.

[2 marks]

b. Interpret the slope coefficient (b) in the context of the study. What does it suggest about the effect of promotional messages on daily meal sales?

[2 marks]

c. Discuss one limitation of using this regression model for decision-making. Consider other factors that might influence the number of meal sets sold.

[2 marks]

QUESTION 3 (20 MARKS)

An animal shelter wants to examine whether the type of housing environment affects the average sleeping time of cats. The shelter currently houses cats in three different settings: an indoor room, an outdoor enclosure, and a mixed environment, where cats spend time both indoors and outdoors.

The shelter manager believes that sleeping patterns may differ depending on the environment. For example, indoor cats may sleep longer due to a quieter setting, while outdoor cats may sleep less because of greater exposure to noise, weather, or activity.

To investigate this, five cats were randomly selected from each environment. Their average sleeping time, measured in hours per day, was recorded as shown below (Table 3).

Table 3

Indoor Room	Outdoor Enclosure	Mixed Environment
15.2	13.8	14.5
14.8	14.1	14.9
15.5	13.5	14.7
15.0	14.0	15.1
14.9	13.9	14.8

Answer the following questions:

a. State the null and alternative hypotheses for this study.

[2 marks]

b. Calculate the sample mean sleeping time for each environment.

[3 marks]

c. Calculate the variance between the sample groups.

[4 marks]

d. Calculate the variance within the sample groups.

[5 marks]

e. At the 5% significance level, test whether there is a significant difference in the mean sleeping time of cats housed in the three environments. State your conclusion in the context of the study.

[6 marks]

QUESTION 4 (15 MARKS)

A café owner is planning future marketing activities and would like to identify whether different promotional campaigns generate different levels of sales revenue. Over a four-week period, the café tested four different promotional campaigns, with each campaign being implemented for one week. Daily sales revenue was recorded during each campaign period.

To compare the effectiveness of the campaigns, the owner summarized the daily sales revenue data. Revenue is measured in hundreds of Malaysian Ringgit (RM). The summary statistics are shown below (Table 4):

Table 4

Campaign	n	Mean Sales, $\bar{x}$	Standard Deviation, s
Campaign A	7	21.0	2.10
Campaign B	7	21.8	2.45
Campaign C	7	20.9	1.95
Campaign D	7	21.5	2.30

Using a significance level of 0.05, perform a one-way ANOVA to test whether the mean daily sales revenue is the same for all four promotional campaigns.

a. State the hypotheses, show all calculations, and provide an appropriate conclusion in the context of the café's marketing strategy.

[13 marks]

b. Based on your findings, what recommendation would you give to the café owner regarding the use of promotional campaigns?

[2 marks]